

Beyond Skewness: Confidence-Guided Rank Fusion for Text Retrieval

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Abstract—Hybrid text retrieval systems commonly combine lexical, learned sparse, and dense retrievers to improve ranking quality across diverse query types. However, standard unsupervised fusion methods typically treat all component retrievers as equally reliable for every query, ignoring query-specific confidence differences. In this work, we extend SkewFuse, an unsupervised confidence-guided rank fusion method originally proposed for visual retrieval, to heterogeneous text retrieval. To address the score-scale mismatch between lexical, sparse, and dense retrievers, we apply per-query score normalization before estimating confidence and performing fusion. Beyond the original skewness-based confidence formulation, we introduce and evaluate alternative confidence estimators based on Top-K Gap, Z-Score, Entropy, and a Hybrid combination. Experiments on ArguAna, NFCorpus, and SciFact show that confidence-guided fusion can outperform standard fusion baselines and that the best confidence estimator varies by dataset. Entropy performs best on ArguAna, Top-K Gap achieves the strongest results on SciFact, and Hybrid remains competitive on NFCorpus. We further analyze model-pair types and show that raw retrieval heterogeneity alone does not guarantee fusion improvement; effective fusion depends on useful complementarity between retrievers. Overall, SkewFuse-Text provides a simple, parameter-light, and fully unsupervised framework for adaptive rank fusion in heterogeneous text retrieval.

Index Terms—text retrieval, rank fusion, hybrid retrieval, confidence estimation, BEIR benchmark, SPLADE, score distribution

I. INTRODUCTION

A central challenge in information retrieval is that no single retriever is optimal across all query types and domains. Lexical methods such as BM25 [1] are effective when queries and documents share surface terms, but they are limited under vocabulary mismatch. Dense retrievers such as E5 [2] capture semantic similarity beyond exact term overlap, while learned sparse models such as SPLADE [3] combine sparse lexical representations with learned expansion. Consequently, modern retrieval systems often combine lexical, sparse, and dense retrievers to obtain more robust rankings.

The effectiveness of such hybrid systems depends on how their outputs are fused. Standard unsupervised fusion methods, including RRF [4], Borda [5], and CombSum [6], usually treat all retrievers as equally reliable for every query. This ignores an important signal: a retriever with a sharply concentrated score distribution is likely more confident than one whose scores are nearly uniform. Therefore, query-specific confidence can provide useful guidance for adaptive fusion.

SkewFuse [7] introduces this idea in multi-view image retrieval by estimating confidence from the skewness of similarity score distributions. In this work, we extend SkewFuse to heterogeneous text retrieval. This extension is non-trivial because lexical, learned sparse, and dense retrievers produce scores with substantially different scales and distributions. Confidence statistics computed on raw scores may therefore reflect model-specific score scale rather than true query-specific certainty. To address this, we apply per-query normalization before confidence estimation and fusion.

We further extend the original skewness-based formulation by evaluating alternative confidence estimators based on Top-K Gap, Z-Score, Entropy, and Hybrid confidence. The resulting framework, *SkewFuse-Text*, remains fully unsupervised and parameter-light while allowing different distributional signals to guide fusion. Experiments on ArguAna [8], NFCorpus [9], and SciFact [10] show that no single confidence estimator dominates all datasets, and that fusion gains depend on useful complementarity where one model’s confident retrievals compensate for another’s failures across query types, rather than raw retrieval heterogeneity alone. In particular, Lexical+Dense pairs provide a stable and practically meaningful setting by combining exact lexical evidence with semantic retrieval.

In summary, this paper makes three contributions: we adapt SkewFuse to heterogeneous text retrieval with per-query score normalization; we compare multiple unsupervised confidence estimators beyond skewness; and we analyze how model-pair type affects fusion effectiveness across three retrieval datasets.

II. BACKGROUND AND RELATED WORK

Text retrieval models can be broadly grouped into lexical, learned sparse, and dense approaches. Lexical models such as BM25 [1] and BM25+ [11] rely on exact term matching and term-frequency signals, making them strong when queries and relevant documents share surface vocabulary. Learned sparse models such as SPLADE [3] preserve vocabulary-level representations while adding learned term expansion, allowing them to bridge lexical matching and semantic retrieval. Dense retrievers such as E5 [2], MiniLM [12], MPNet [13], and MultiQA [14] encode queries and documents into continuous vector spaces and retrieve documents by semantic similarity. Since these model families capture different retrieval signals, hybrid retrieval systems often combine them to improve robustness across domains [15], [16].

Unsupervised rank fusion methods combine outputs from multiple retrievers without requiring labeled training data. Score-based methods such as CombSum and CombMNZ [6] aggregate normalized retrieval scores, while rank-based methods such as Borda [5], Dowdall [17], and RRF [4] rely on document positions rather than raw scores. Rank-based methods are naturally robust to score-scale mismatch, which is important when combining lexical, sparse, and dense retrievers. Condorcet fusion [18] selects the document that wins pairwise comparisons against all other candidates, providing another rank-based alternative. ISR and its variants [19] introduce score normalization to mitigate distribution biases across retrieval methods. However, these methods usually treat all retrievers as equally reliable for every query and do not explicitly model query-specific confidence.

Confidence-aware fusion methods attempt to estimate the reliability of each retrieval output before aggregation. SegFuse [20] and CombGMNZ [21] use ranking or score-distribution properties to guide fusion, while supervised methods [22], [23] learn fusion weights from relevance judgments. SkewFuse [7] is the closest work to ours: it estimates query-specific confidence using the skewness of similarity score distributions and applies confidence-weighted rank fusion in visual retrieval. We extend this idea to heterogeneous text retrieval, where model families produce scores on different scales, and compare additional confidence estimators based on Top-K Gap, Z-Score, Entropy, and Hybrid confidence.

III. METHODOLOGY

A. Overview

We propose an unsupervised confidence-guided fusion framework for heterogeneous text retrieval. Given a query q , each retriever m returns a ranked list of candidate documents with scores:

$$R_q^m = \{(d_i^m, s_i^m)\}_{i=1}^L, \quad (1)$$

where d_i^m is the document at rank i , s_i^m is its retrieval score, and L is the number of retrieved candidates. The goal is to combine the outputs of multiple retrievers without supervised training labels or learned fusion weights.

The proposed framework consists of three steps. First, retrieval scores are normalized per query to make heterogeneous model outputs comparable. Second, each model's score distribution is converted into a query-specific confidence score. Third, normalized scores are aggregated using a confidence-weighted rank fusion rule.

B. Per-Query Score Normalization

A direct application of SkewFuse to text retrieval is challenging because different retriever families produce scores on different scales. For example, BM25 and SPLADE generate sparse lexical scores that are often unbounded, while dense encoders such as E5 or MiniLM typically produce cosine or dot-product similarities within a narrower range. Computing confidence statistics on raw scores would therefore mix query-specific certainty with model-specific score-scale effects.

To address this issue, we apply min-max normalization separately for each query and model. Let

$$S_q^m = \{s_1^m, s_2^m, \dots, s_L^m\} \quad (2)$$

be the score list returned by model m for query q . Each score is normalized as:

$$\tilde{s}_i^m = \frac{s_i^m - \min S_q^m}{\max S_q^m - \min S_q^m + \epsilon}, \quad (3)$$

where ϵ is a small constant for numerical stability. This maps scores to a comparable range while preserving the ranking order within each model. Confidence estimation is then performed on the normalized score list $\tilde{S}_q^m = \{\tilde{s}_1^m, \dots, \tilde{s}_L^m\}$.

C. Confidence Estimation

We estimate model confidence from the shape of the normalized score distribution. The intuition is that a model is more confident when it assigns much higher scores to a small number of top-ranked documents, and less confident when its scores are flat or weakly separated. We evaluate five confidence variants: the original skewness-based SkewFuse confidence and four extensions based on Top-K Gap, Z-Score, Entropy, and Hybrid confidence.

1) *Skewness Confidence*: The original SkewFuse confidence uses the skewness of the score distribution. A right-skewed distribution indicates that a few documents receive substantially higher scores than the rest, suggesting stronger model confidence. Let $g(\tilde{S}_q^m)$ denote the sample skewness of the normalized score list. The confidence is defined as:

$$C_{\text{skew}}(\tilde{S}_q^m) = \frac{\max(0, g(\tilde{S}_q^m))}{\max(0, g(\tilde{S}_q^m)) + K}, \quad (4)$$

where $K > 0$ controls the sharpness of the confidence scaling. This corresponds to the original SkewFuse formulation [7].

2) *Top-K Gap Confidence*: Top-K Gap confidence measures the separation between the highest-scoring document and the h -th ranked document. Let $\tilde{s}_{(1)}$ and $\tilde{s}_{(h)}$ denote the first and h -th highest normalized scores, respectively. We define:

$$\Delta_h = \tilde{s}_{(1)} - \tilde{s}_{(h)}, \quad (5)$$

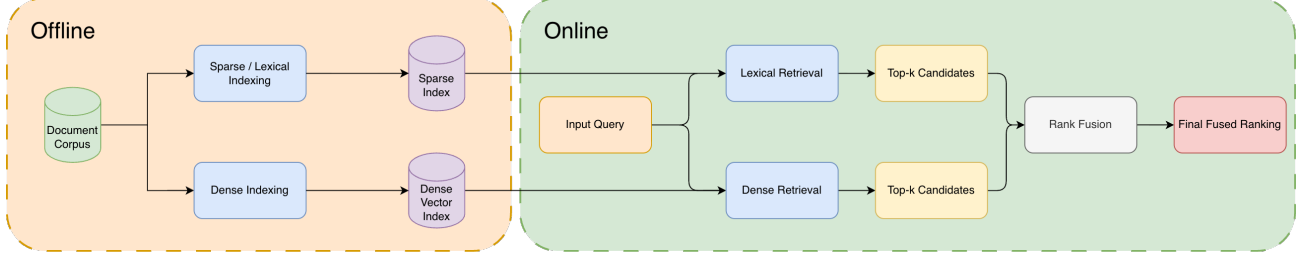


Fig. 1. Overview of the SkewFuse-Text pipeline. For each query, heterogeneous retrievers produce ranked candidate lists with model-specific scores. Scores are first normalized per query, then converted into confidence values using distributional statistics. The final ranking is obtained through confidence-guided rank fusion.

$$C_{\text{gap}}(\tilde{S}_q^m) = \frac{\Delta_h}{\Delta_h + K}. \quad (6)$$

A larger gap indicates that the model strongly prefers its top candidate over the remaining candidates. In our experiments, we use $h = 5$.

3) *Z-Score Confidence*: Z-Score confidence measures how far the maximum score is from the mean score in units of standard deviation:

$$z_{\text{max}} = \frac{\max \tilde{S}_q^m - \mu_{\tilde{S}_q^m}}{\sigma_{\tilde{S}_q^m} + \epsilon}, \quad (7)$$

$$C_z(\tilde{S}_q^m) = \frac{\max(0, z_{\text{max}})}{\max(0, z_{\text{max}}) + K}. \quad (8)$$

This metric captures whether the top-ranked document is statistically separated from the rest of the retrieved list.

4) *Entropy Confidence*: Entropy confidence measures how concentrated the normalized scores are. We first convert normalized scores into a probability distribution:

$$p_i = \frac{\tilde{s}_i^m + \epsilon}{\sum_j (\tilde{s}_j^m + \epsilon)}. \quad (9)$$

The normalized entropy is then computed as:

$$H(\tilde{S}_q^m) = - \sum_i p_i \log p_i, \quad (10)$$

$$C_{\text{ent}}(\tilde{S}_q^m) = 1 - \frac{H(\tilde{S}_q^m)}{\log |\tilde{S}_q^m|}. \quad (11)$$

Lower entropy indicates that score mass is concentrated on fewer documents, which corresponds to higher retrieval confidence.

5) *Hybrid Confidence*: Finally, the Hybrid variant combines all confidence estimators using a simple average:

$$C_{\text{hyb}}(\tilde{S}_q^m) = \frac{C_{\text{skew}} + C_{\text{gap}} + C_z + C_{\text{ent}}}{4}. \quad (12)$$

This variant provides a robust default when the most suitable confidence metric is unknown.

D. Confidence-Guided Rank Fusion

After computing a confidence score for each model and query, we aggregate candidate documents using a confidence-weighted rank fusion rule. For a document d , the final fused score is:

$$F_q(d) = \sum_m \sum_{i=1}^L \mathbf{1}(d = d_i^m) \cdot \frac{C_q^m \cdot \tilde{s}_i^m}{i}, \quad (13)$$

where C_q^m is the confidence of model m for query q , $\tilde{s}_i^m$ is the normalized score of the document at rank i , and the $1/i$ term applies a Dowdall-style positional discount.

This formulation gives higher contribution to documents that are highly ranked, have high normalized scores, and come from models with higher query-specific confidence. Documents retrieved by multiple models accumulate contributions from each list. The final output is obtained by sorting all candidate documents according to $F_q(d)$.

The original SkewFuse method is recovered by setting $C_q^m = C_{\text{skew}}(\tilde{S}_q^m)$. The proposed SkewFuse-Text variants replace this confidence term with Top-K Gap, Z-Score, Entropy, or Hybrid confidence while keeping the same unsupervised fusion rule.

IV. EXPERIMENTAL SETUP

A. Datasets

We evaluate on three publicly available BEIR [24] datasets with diverse retrieval characteristics (Table I). *SciFact* [10] requires matching natural language claims against biomedical abstracts. *NFCorpus* [9] is known to challenge neural models due to vocabulary mismatch between lay-language queries and technical medical documents. *ArguAna* [8] requires counter-argument retrieval, making it primarily a semantic similarity task.

TABLE I
BEIR DATASET CHARACTERISTICS

Dataset	Domain	Corpus	Queries
SciFact [10]	Scientific claim verification	5,183	300
NFCorpus [9]	Medical / nutrition IR	3,633	323
ArguAna [8]	Argument counter-retrieval	8,674	1,406

B. Models and Baselines

We evaluate seven retrieval models spanning three paradigms. *Lexical*: BM25 and BM25+. *Learned sparse*: SPLADE. *Dense*: E5, MPNet, MiniLM, and MultiQA. All models retrieve top-100 candidates per query.

Fusion baselines are: RRF, CombSum, CombMNZ, Borda, Condorcet, ISR and Dowdall, applied to all pairwise lexical-dense model combinations. SkewFuse-Text is evaluated with the original skewness-based confidence estimator and four alternative confidence variants, using different K values. We observe that performance is stable across $K \in [0.1, 10.0]$, with negligible variation in nDCG@10; results degrade only at extreme values ($K > 100$) where confidence scores collapse toward zero.

C. Evaluation Protocol

We report nDCG@3, nDCG@5 and nDCG@10 on the official BEIR test splits using the ranx library [25]. We distinguish *global-level wins* (SkewFuse outperforms all single models for the dataset) from *pair-level wins* (SkewFuse outperforms the stronger constituent model and the best standard fusion baseline on the same pair). Global-level wins are our primary criterion.

V. RESULTS

A. Single-Model Performance

Table II reports the nDCG@10 performance of individual retrieval models across the three datasets. The results show that no single retriever dominates all retrieval settings. SPLADE achieves the strongest performance on ArguAna, indicating the effectiveness of learned sparse expansion for argument retrieval. In contrast, E5 obtains the best results on NFCorpus and SciFact, suggesting that dense semantic representations are particularly effective for medical and scientific claim retrieval. Lexical baselines such as BM25 and BM25+ generally trail the learned sparse and dense models, although BM25+ remains competitive on ArguAna.

TABLE II
SINGLE-MODEL NDCG@10 PERFORMANCE ACROSS DATASETS. THE HIGHEST SCORE IN EACH DATASET IS SHOWN IN BOLD.

Method	ArguAna	NFCorpus	SciFact
SPLADE [3]	0.394	0.332	0.633
MiniLM [12]	0.370	0.316	0.645
MultiQA [14]	0.366	0.319	0.593
MPNet [13]	0.362	0.333	0.656
E5 [2]	0.331	0.353	0.711
BM25+ [11]	0.327	0.272	0.566
BM25 [1]	0.263	0.268	0.552

B. Model-Pair Type Analysis

Before conducting the detailed comparison of fusion methods, we analyze how different model-pair categories behave. We group model pairs into three categories: Lexical+Lexical,

Dense+Dense, and Lexical+Dense. Figure 2 reports the retrieval heterogeneity of these groups, measured as one minus the top- k Jaccard overlap between the retrieved document sets. Figure 3 reports the maximum fusion gain over the best single model for each pair type.

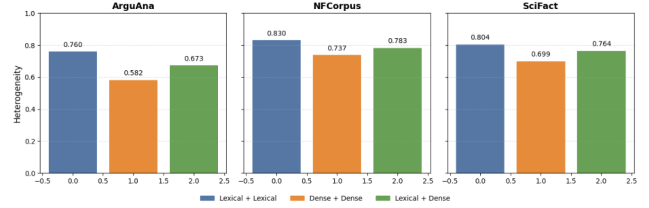


Fig. 2. Retrieval heterogeneity across model-pair types. Heterogeneity is measured as one minus the top- k Jaccard overlap between retrieved document sets. Lexical+Lexical pairs exhibit the highest disagreement, while Dense+Dense pairs are less heterogeneous.

The results show that raw heterogeneity alone does not guarantee effective fusion. Lexical+Lexical pairs exhibit the highest retrieval heterogeneity across all datasets, but this does not consistently translate into performance gains. In fact, Lexical+Lexical fusion substantially degrades performance on NFCorpus and SciFact. Dense+Dense pairs show lower heterogeneity and achieve positive gains on NFCorpus and SciFact, but they do not expose the lexical-semantic mismatch that motivates hybrid retrieval. Lexical+Dense pairs provide a more balanced setting: they maintain moderate-to-high heterogeneity and yield positive fusion gains across all datasets. This motivates using Lexical+Dense combinations as the primary setting for detailed comparison of confidence-guided fusion methods.

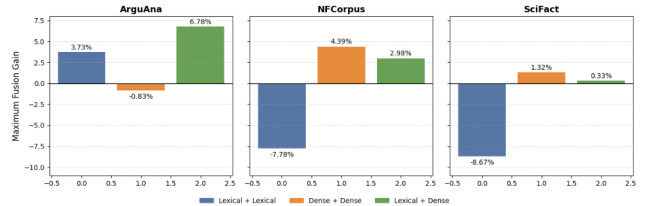


Fig. 3. Maximum fusion gain over the best single model across model-pair types. Positive values indicate that the best fusion result improves over the strongest individual retriever in the pair. Lexical+Dense pairs provide positive gains across all datasets, while Lexical+Lexical pairs fail on NFCorpus and SciFact.

C. Detailed Fusion Results

Table III compares standard rank-fusion baselines with the SkewFuse-family methods on the selected hybrid pair, E5+SPLADE. We select this pair because it represents a strong Lexical/Sparse+Dense setting: E5 provides dense semantic retrieval, while SPLADE provides learned sparse lexical expansion. This pair therefore captures the score-scale and representation mismatch that motivates SkewFuse-Text.

The results demonstrate that confidence-guided fusion can improve over conventional aggregation methods, but the best

TABLE III
NDCG PERFORMANCE OF STANDARD FUSION BASELINES AND SKEWFUSE-FAMILY METHODS ACROSS DATASETS. BEST RESULTS ARE SHOWN IN BOLD AND SECOND-BEST RESULTS ARE UNDERLINED.

Method	ArguAna			NFCorpus			SciFact		
	nDCG@3	nDCG@5	nDCG@10	nDCG@3	nDCG@5	nDCG@10	nDCG@3	nDCG@5	nDCG@10
BordaFuse [5]	0.280	0.346	0.401	0.420	0.392	0.353	0.650	0.668	0.690
CombMNZ [6]	0.277	0.344	0.402	0.423	<u>0.396</u>	0.360	0.657	0.679	0.702
CombSum [6]	0.276	0.342	0.397	<u>0.427</u>	0.397	<u>0.363</u>	0.657	0.681	0.704
Condorcet [18]	0.243	0.304	0.353	0.397	0.370	0.335	0.617	0.644	0.653
ISR [19]	0.264	0.334	0.395	0.422	0.393	0.361	<u>0.662</u>	<u>0.684</u>	0.701
RRF [4]	0.283	0.347	<u>0.403</u>	0.418	0.392	0.354	0.646	0.668	0.693
Dowdall [17]	0.272	0.339	0.399	0.420	0.395	0.361	0.659	<u>0.684</u>	0.701
SkewFuse [7]	0.280	0.344	0.400	0.423	0.395	0.364	0.661	0.683	<u>0.705</u>
SkewFuse-Top-K Gap	0.281	0.345	0.401	0.423	0.395	0.362	0.664	0.688	0.713
SkewFuse-Z-Score	0.277	0.342	0.398	0.428	0.397	0.364	0.659	0.681	<u>0.705</u>
SkewFuse-Entropy	0.287	0.352	0.408	0.423	0.394	0.361	0.659	0.683	0.704
SkewFuse-Hybrid	0.280	0.345	0.401	0.425	0.397	0.364	0.657	0.681	<u>0.705</u>

confidence estimator varies across datasets. On ArguAna, SkewFuse-Entropy achieves the strongest performance across all reported cutoffs, reaching an nDCG@10 of 0.408. This suggests that entropy-based concentration is useful in the argument retrieval setting, where semantic ambiguity can make distributional confidence informative. On NFCorpus, SkewFuse, SkewFuse-Z-Score, and SkewFuse-Hybrid obtain the best nDCG@10 score of 0.364, while SkewFuse-Z-Score also achieves the highest nDCG@3. On SciFact, SkewFuse-Top-K Gap performs best, achieving an nDCG@10 of 0.713 and outperforming both the original SkewFuse and all standard fusion baselines.

Overall, these results indicate that the original skewness-based SkewFuse remains a strong baseline, while alternative confidence estimators provide dataset-specific gains. Entropy is strongest on ArguAna, Top-K Gap is strongest on SciFact, and Z-Score/Hybrid are competitive on NFCorpus. This supports the motivation for extending SkewFuse from a single skewness-based confidence estimator into a broader family of distributional confidence estimators.

D. Robustness Across Model-Pair Combinations

To evaluate the robustness of the proposed confidence estimators, we further average each SkewFuse-family method across all model-pair combinations. Table IV reports the resulting average nDCG@10 scores.

TABLE IV
AVERAGE NDCG@10 OF SKEWFUSE-FAMILY METHODS ACROSS ALL MODEL-PAIR COMBINATIONS. BEST RESULT FOR EACH DATASET IS SHOWN IN BOLD.

Method	ArguAna	NFCorpus	SciFact
SkewFuse	0.381	0.342	0.672
SkewFuse-Top-K Gap	0.382	0.340	0.673
SkewFuse-Z-Score	0.379	0.341	0.671
SkewFuse-Entropy	0.381	0.340	0.669
SkewFuse-Hybrid	0.381	0.342	0.672

The results show that no single confidence estimator dominates all datasets. SkewFuse-Top-K Gap achieves the best average performance on ArguAna and SciFact, while SkewFuse-Hybrid performs slightly better on NFCorpus. The differences among the top methods are small, indicating that the SkewFuse-family methods are generally stable across different model-pair choices. Nevertheless, Top-K Gap and Hybrid emerge as the most robust alternatives: Top-K Gap provides the best average performance on two datasets, while Hybrid remains highly competitive and achieves the best average score on NFCorpus.

VI. CONCLUSION AND FUTURE WORK

This work extends SkewFuse from visual retrieval to heterogeneous text retrieval as a fully unsupervised confidence-guided rank fusion framework. The method estimates query-specific model confidence from retrieval score distributions and applies per-query normalization to address score-scale differences between lexical, learned sparse, and dense retrievers.

We evaluated the original skewness-based SkewFuse together with Top-K Gap, Z-Score, Entropy, and Hybrid confidence variants on ArguAna, NFCorpus, and SciFact. The results show that no single confidence estimator dominates all datasets: Entropy performs best on ArguAna, Top-K Gap is strongest on SciFact, and Hybrid remains competitive on NFCorpus. Our pair-type analysis also shows that raw retrieval heterogeneity alone is not sufficient for effective fusion; Lexical+Dense pairs provide a more stable and practically meaningful balance between lexical matching and semantic retrieval.

Overall, the SkewFuse-family methods provide a simple, parameter-light, and unsupervised alternative to traditional fusion methods that treat all retrievers uniformly. Future work will focus on adaptive weighting strategies, including weighted combinations of multiple confidence estimators instead of simple averaging, weighted combinations of different fusion methodologies such as RRF, Dowdall, CombSum, and SkewFuse variants, and extensions from pairwise fusion

to confidence-weighted fusion over more than two retrieval models.

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